

class09

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Background

Today we are taking a detour to analyze a fun dataset (that we have more intrinsic insight into) with the most useful analysis method we have learned thus far - Principal Component Analysis (PCA)

Data Import

The data is all related to halloween candy and is from the 538 website.

```
# Store the file name as a variable
candy_file <- "candy-data.txt"
candy <- read.csv(candy_file, row.names = 1)

# View the first few rows of the dataset
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

```
# install.packages("ggplot2")
# install.packages("skimr")
# install.packages("corrplot")
# install.packages("ggrepel")
```

Q1. How many different candy types are in this dataset?

```
# Get number of rows (# of rows = candy types)
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset?

```
# Sum up all rows where fruity is equal to 1, since 1=yes, 0=no
sum(candy$fruity)
```

```
[1] 38
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is its winpercent value?

```
# Get winpercent for favorite candy
candy["Reese's Peanut Butter cup", ]$winpercent
```

```
[1] 84.18029
```

Q4. What is the winpercent value for “Kit Kat”?

```
# Get winpercent for Kit Kat  
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
# Get winpercent for Tootsie Roll Snack Bars  
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

```
# Load the skimr package (only need to install once)  
library(skimr)  
  
# Get a summary of the dataset  
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Yes, the winpercent column is on a different scale because it ranges from 0–100, while most other variables are binary (for example, 0 or 1).

Q7. What do you think a zero and one represent for the candy\$chocolate column?

```
# View the values in chocolate column
table(candy$chocolate)
```

```
0 1
48 37
```

A value of 1 means the candy contains chocolate, and 0 means it does not.

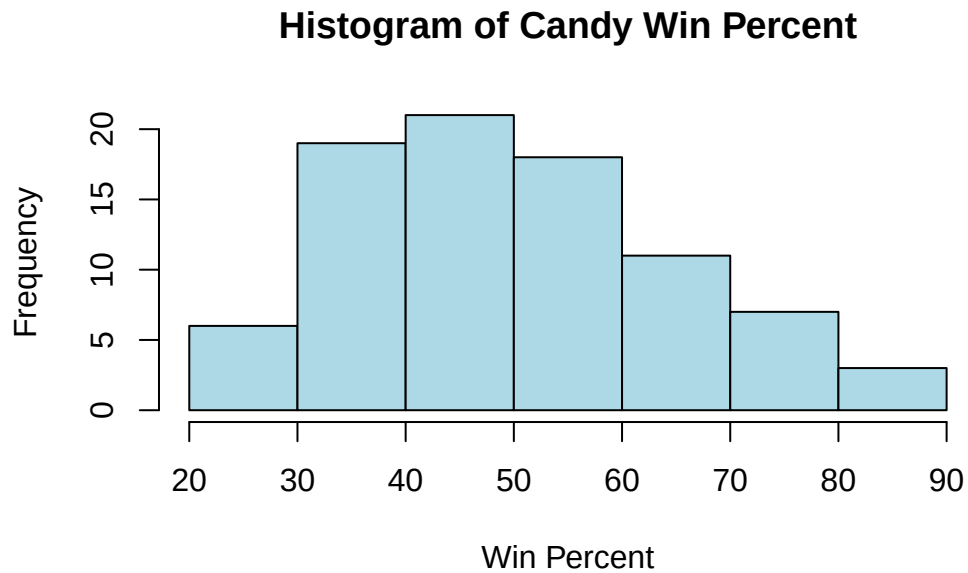
Exploratory analysis

Q8. Plot a histogram of winpercent values using both base R and ggplot2.

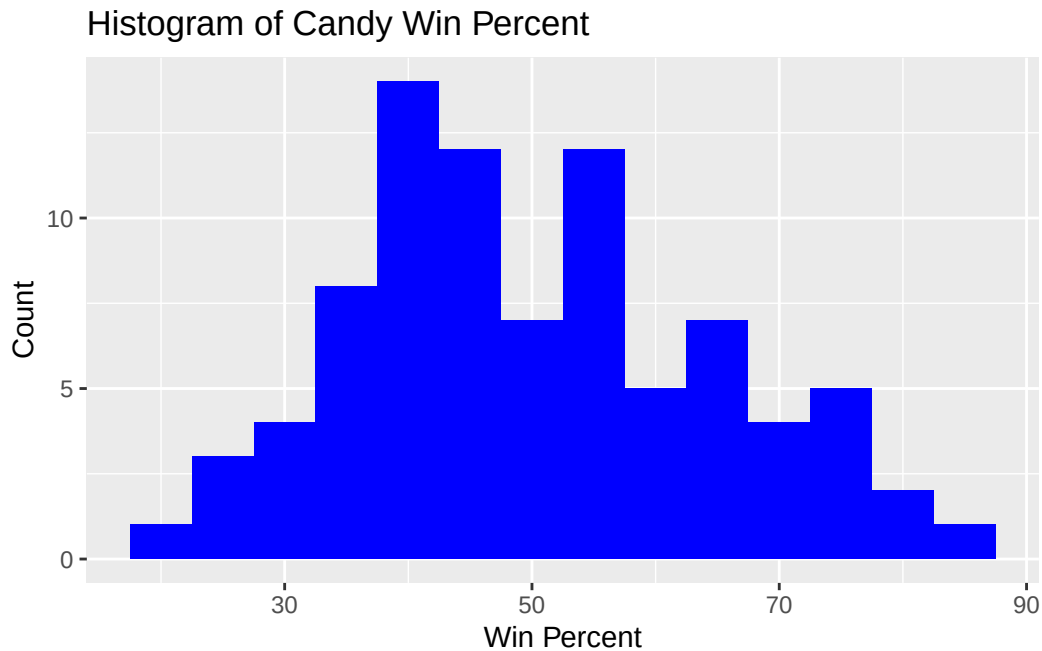
```
# Load ggplot2 (install once, then comment it out)
# install.packages("ggplot2")
library(ggplot2)

# Histogram using base R
hist(candy$winpercent,
```

```
main = "Histogram of Candy Win Percent",  
xlab = "Win Percent",  
col = "lightblue")
```



```
# Create histogram using ggplot  
ggplot(candy, aes(x = winpercent)) +  
  geom_histogram(binwidth = 5, fill = "blue") +  
  labs(title = "Histogram of Candy Win Percent",  
        x = "Win Percent",  
        y = "Count")
```



Q9. Is the distribution of winpercent values symmetrical?

No, the distribution is not perfectly symmetrical. It appears slightly skewed to the right.

Q10. Is the center of the distribution above or below 50%?

The center of the distribution is slightly above 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
# Average winpercent for chocolate candies  
mean(candy$winpercent[as.logical(candy$chocolate)])
```

```
[1] 60.92153
```

```
# Average winpercent for fruity candies  
mean(candy$winpercent[as.logical(candy$fruity)])
```

```
[1] 44.11974
```

On average, chocolate candies are ranked higher than fruity candies. The mean winpercent for chocolate candies is substantially higher than for fruity candies.

Q12. Is this difference statistically significant?

```
# Compare chocolate vs fruity winpercent using t-test
t.test(candy$winpercent[as.logical(candy$chocolate)],
       candy$winpercent[as.logical(candy$fruity)])
```

Welch Two Sample t-test

```
data: candy$winpercent[as.logical(candy$chocolate)] and candy$winpercent[as.logical(candy$fruity)]
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Yes, the difference is statistically significant, as the p-value of 2.87×10^{-8} is far below 0.05, indicating a highly significant difference in popularity between chocolate and fruity candies.

Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

```
# Sort candy from lowest winpercent to highest winpercent
least_liked <- candy[order(candy$winpercent), ]

# Show the first 5 rows, which are the least liked candies
head(least_liked, n = 5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Nik L Nip	0	1	0	0	0
Boston Baked Beans	0	0	0	1	0
Chiclets	0	1	0	0	0
Super Bubble	0	1	0	0	0
Jawbusters	0	1	0	0	0

	crispedricewafer	hard bar	pluribus	sugarpercent	pricepercent	
Nik L Nip	0	0	0	1	0.197	0.976
Boston Baked Beans	0	0	0	1	0.313	0.511

Chiclets	0	0	0	1	0.046	0.325
Super Bubble	0	0	0	0	0.162	0.116
Jawbusters	0	1	0	1	0.093	0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

Q14. What are the top 5 all time favorite candy types out of this set?

```
# Sort candy from highest winpercent to lowest winpercent
most_liked <- candy[order(candy$winpercent, decreasing = TRUE), ]

# Show the first 5 rows, which are the most liked candies
head(most_liked, n = 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1

	crisped	rice	wafer	hard	bar	pluribus	sugar
Reese's Peanut Butter cup		0	0	0		0	0.720
Reese's Miniatures		0	0	0		0	0.034
Twix		1	0	1		0	0.546
Kit Kat		1	0	1		0	0.313
Snickers		0	0	1		0	0.546

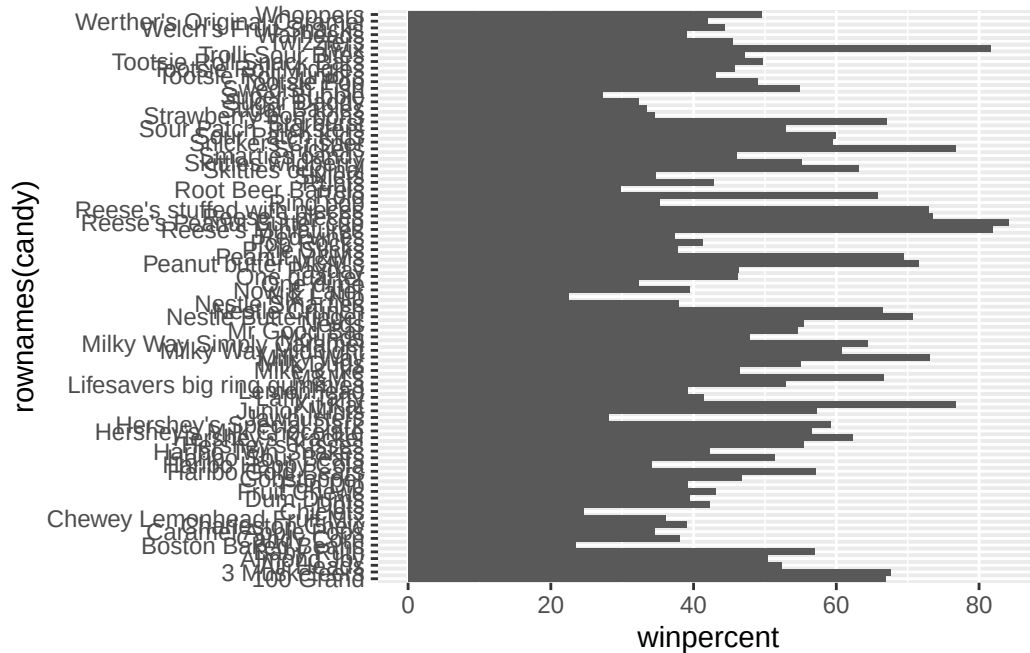
	price	percent	winpercent
Reese's Peanut Butter cup	0.651		84.18029
Reese's Miniatures	0.279		81.86626
Twix	0.906		81.64291
Kit Kat	0.511		76.76860
Snickers	0.651		76.67378

Q15. Make a first barplot of candy ranking based on winpercent values.

```
# Load ggplot2 for plotting
library(ggplot2)
```

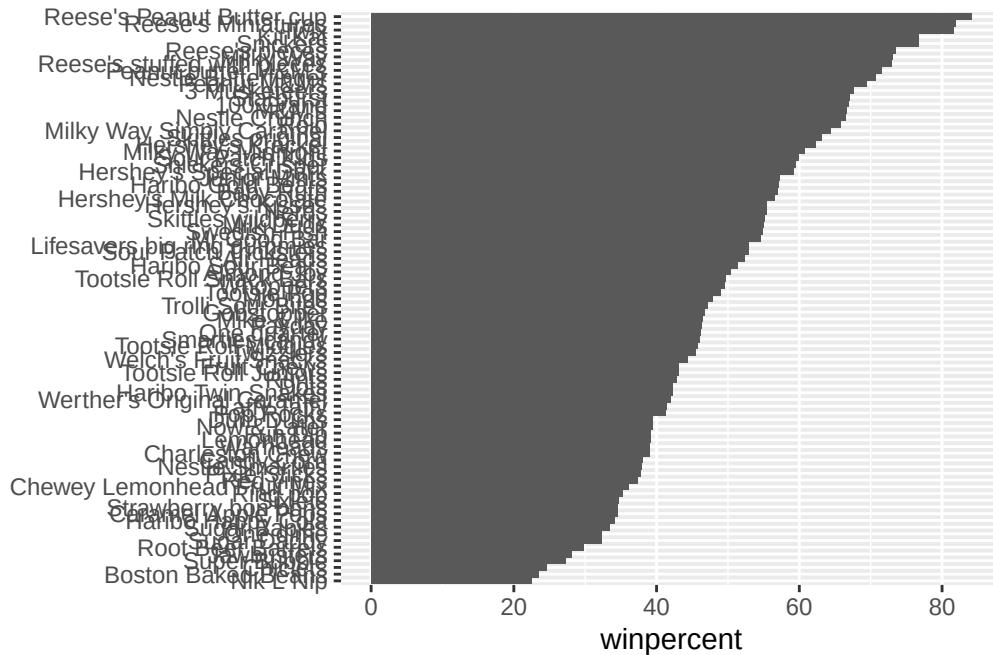


```
# Make a basic barplot of winpercent for each candy
ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```



Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

```
# Make a better plot with candies sorted by winpercent
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col() +
  ylab("")
```



Q17. What is the worst ranked chocolate candy?

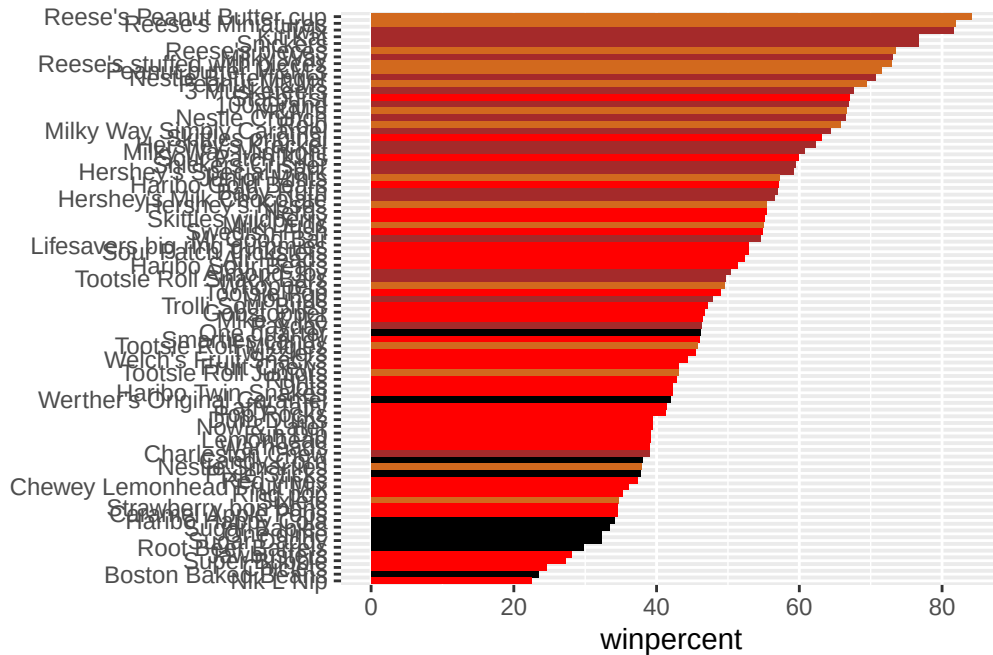
```
# Making a vector of "black" for every candy
my_cols <- rep("black", nrow(candy))

# Changing color to "chocolate" for candies that contain chocolate
my_cols[as.logical(candy$chocolate)] <- "chocolate"

# Changing color to "brown" for candies that are bars
my_cols[as.logical(candy$bar)] <- "brown"

# Changing color to "red" for fruity candies
my_cols[as.logical(candy$fruity)] <- "red"
```

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col(fill=my_cols) +
  ylab("")
```



```
# Get only chocolate candies
choc <- candy[as.logical(candy$chocolate), ]

# Sort from lowest to highest winpercent
choc_sorted <- choc[order(choc$winpercent), ]

# Show only the worst one
head(choc_sorted, 1)
```

```
      chocolate fruity caramel peanutyalmondy nougat crispedricewafer hard
Sixlets      1      0      0      0      0      0      0
      bar pluribus sugarpercent pricepercent winpercent
Sixlets      0      1      0.22      0.081      34.722
```

Sixlets are the worst ranked chocolate candy.

Q18. What is the best ranked fruity candy?

```
# Get only fruity candies
fruity <- candy[as.logical(candy$fruity), ]

# Sort from highest to lowest winpercent
```

```
fruity_desc <- fruity[order(fruity$winpercent, decreasing = TRUE), ]

# Show best one
head(fruity_desc, 1)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer	hard
Starburst	0	1	0	0	0	0	0

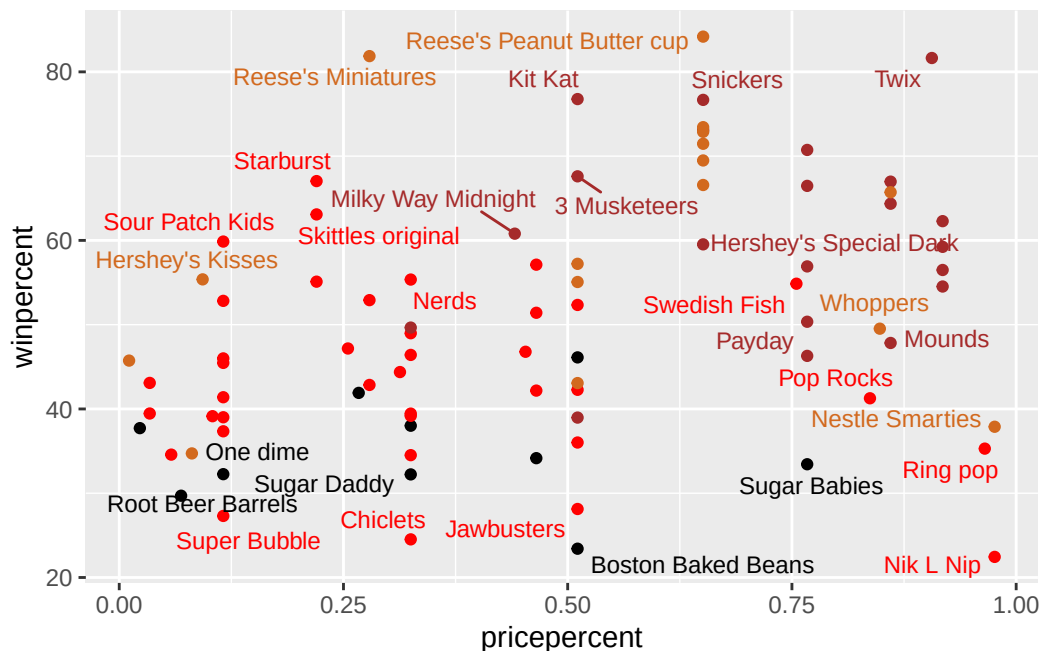
	bar	pluribus	sugarpercent	pricepercent	winpercent
Starburst	0	1	0.151	0.22	67.03763

Starburst is the best ranked fruity candy.

Taking a look at pricepercent

```
# install.packages("ggrepel") and load it
library(ggrepel)

# Make a scatter plot comparing pricepercent and winpercent
ggplot(candy) +
  aes(pricepercent, winpercent, label = rownames(candy)) +
  geom_point(col = my_cols) +
  geom_text_repel(col = my_cols, size = 3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

```
# Look at candies sorted by highest winpercent
head(candy[order(candy$winpercent, decreasing = TRUE), c("pricepercent", "winpercent")], 10)
```

	pricepercent	winpercent
Reese's Peanut Butter cup	0.651	84.18029
Reese's Miniatures	0.279	81.86626
Twix	0.906	81.64291
Kit Kat	0.511	76.76860
Snickers	0.651	76.67378
Reese's pieces	0.651	73.43499
Milky Way	0.651	73.09956
Reese's stuffed with pieces	0.651	72.88790
Peanut butter M&M's	0.651	71.46505
Nestle Butterfinger	0.767	70.73564

Reese's Miniatures offers the best value for money because it has a very high winpercent of around 81.87 percent while also having a relatively low pricepercent of 0.279, making it both popular and inexpensive.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
# Order candies from most expensive to least expensive
ord <- order(candy$pricepercent, decreasing = TRUE)

# Show the top 5 most expensive candies with pricepercent and winpercent
head(candy[ord, c(11, 12)], n = 5)
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

The top 5 most expensive candies are Nik L Nip, Nestle Smarties, Ring Pop, Hershey's Krackel, and Hershey's Milk Chocolate. Nik L Nip is the least popular because it has the lowest winpercent of around 22.4.

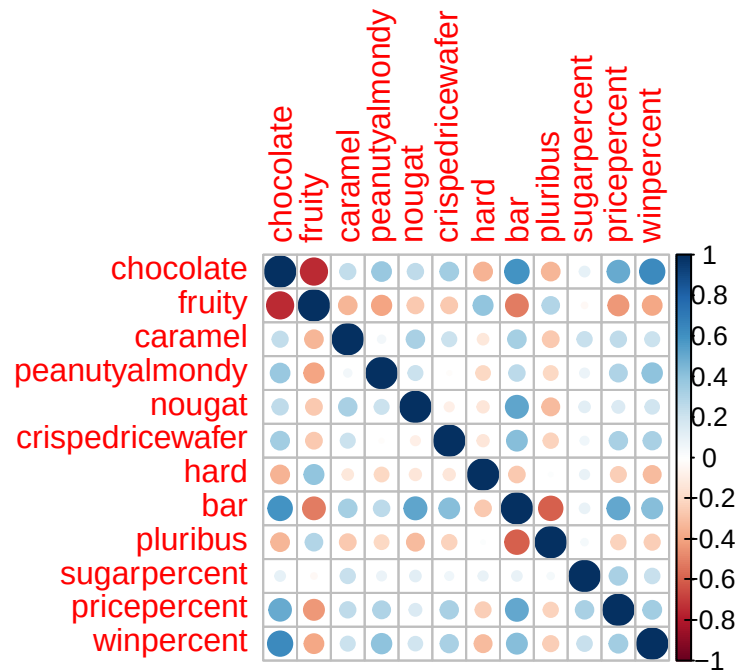
Exploring the correlation structure

```
# install.packages("corrplot") then run it
library(corrplot)
```

corrplot 0.95 loaded

```
# Find correlation matrix for all variables
cij <- cor(candy)

# Plotting the correlation matrix
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

The variables chocolate and fruity are strongly anti-correlated.

Q23. Use your corplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

The variables chocolate and fruity will have the largest contributions to PC1. Chocolate is positively correlated with variables like bar, pricepercent, and winpercent, while fruity is negatively correlated with these. There will be a strong contrast, which will drive the main separation in the PCA.

Principal Component Analysis (PCA)

```
# PCA on the candy dataset with scaling
pca <- prcomp(candy, scale = TRUE)

# Summary of PCA results
summary(pca)
```

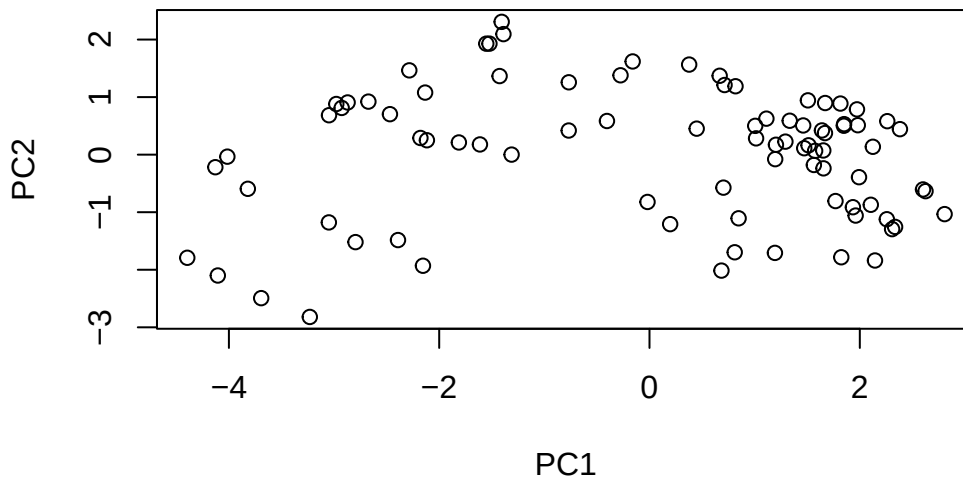
Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

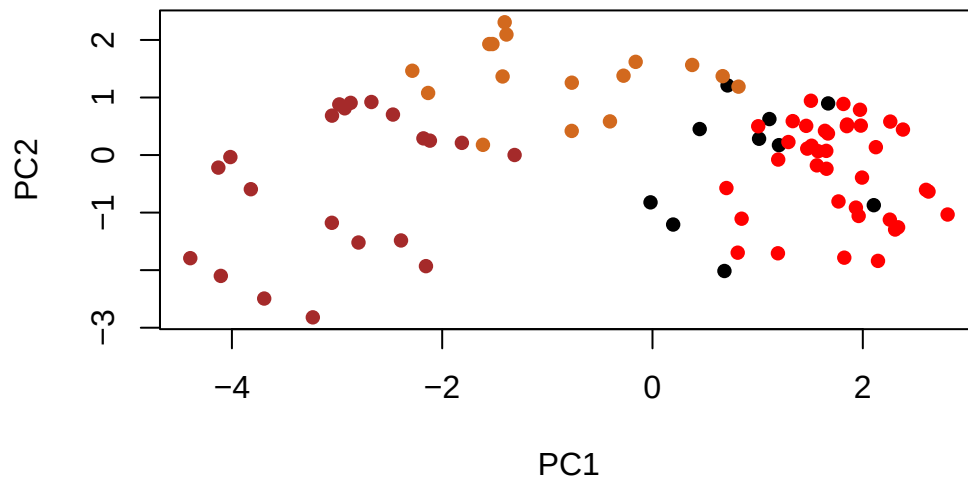
```
# Plot of PC1 vs PC2
```

```
plot(pca$x[,1:2])
```



```
# With color and filled points
```

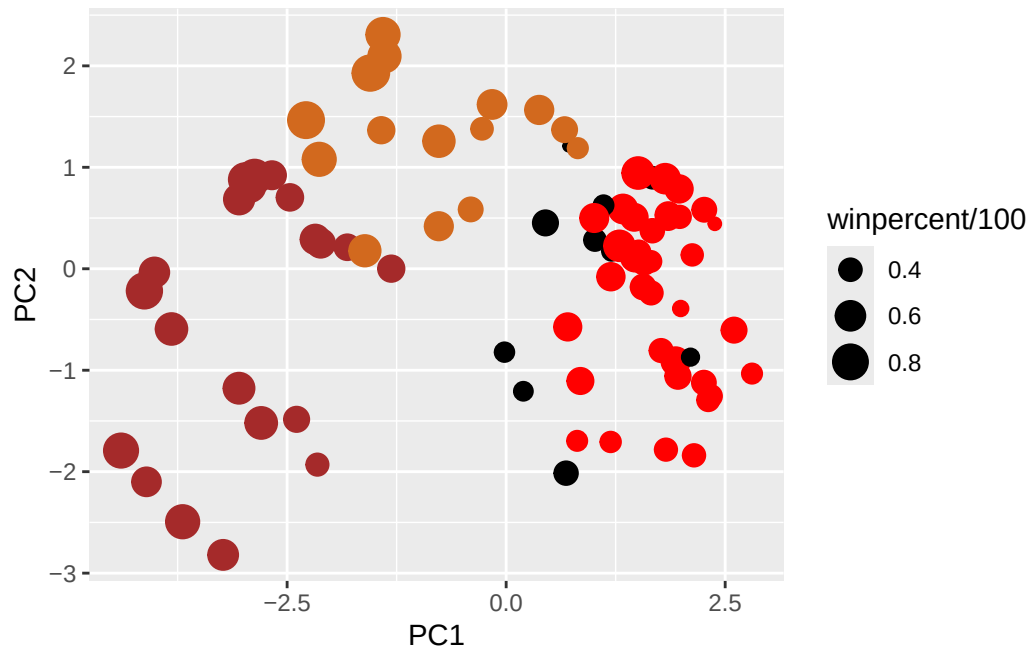
```
plot(pca$x[,1:2], col = my_cols, pch = 16)
```

```
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])

# Create PCA scatter plot
p <- ggplot(my_data) +
  aes(x = PC1, y = PC2,                                # use PC1 and PC2
      size = winpercent/100,
      text = rownames(my_data),
      label = rownames(my_data)) +
  geom_point(col = my_cols)                             # color by candy type

p
```

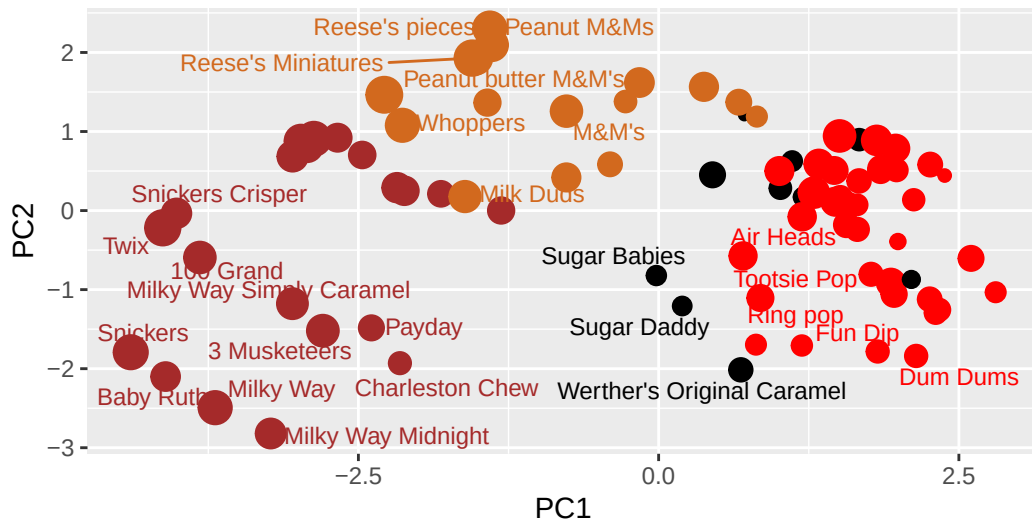


```
library(ggrepel)

p + geom_text_repel(size = 3.3, col = my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title = "Halloween Candy PCA Space",
        subtitle = "Colored by type: chocolate bar (dark brown), chocolate other (light brown)",
        caption = "Data from 538")
```

Halloween Candy PCA Space

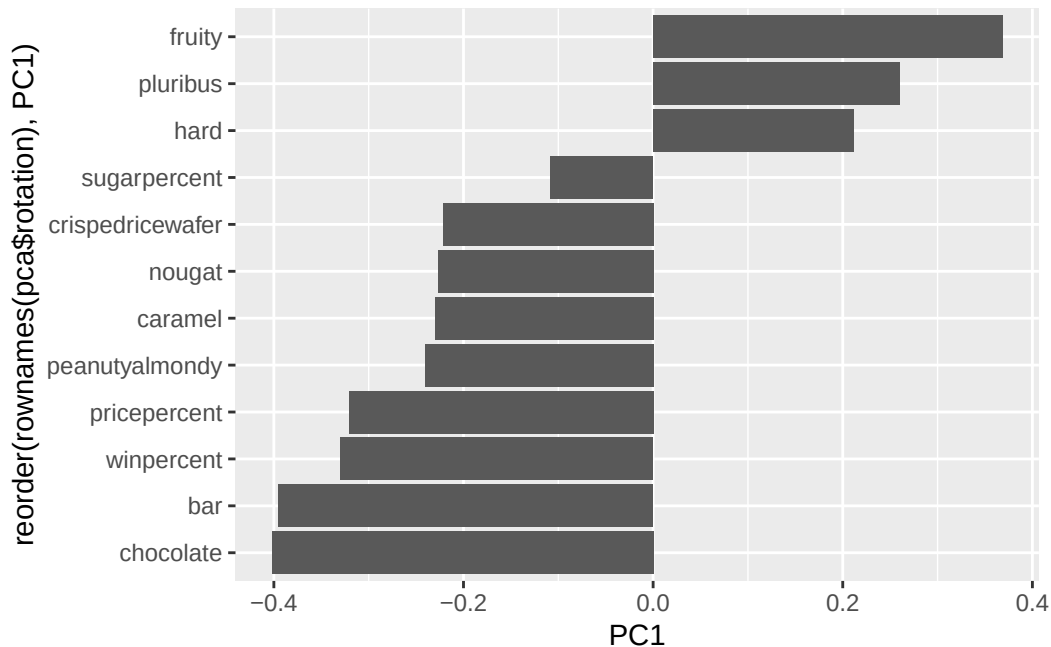
Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

```
# Plot PCA loadings (PC1 values vs variable names)
ggplot(pca$rotation) +
  aes(PC1, reorder(rownames(pca$rotation), PC1)) +
  geom_col()
```



The variables `fruity`, `pluribus`, and `hard` contribute strongly in the positive direction of PC1. This shows that candies that are fruity, hard, and that come in multiple pieces tend to cluster together along this axis. This makes sense because these variables are negatively correlated with `chocolate`, `bar`, and `winpercent`, which were observed earlier in the correlation matrix. This means that PC1 represents a separation between fruity-type candies and chocolate-based candies, consistent with patterns seen in the `corrplot`.

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

A “winning” candy should be chocolate-based, a bar, and relatively higher in price, and these candies generally have higher `winpercent` values. From the exploratory analysis, chocolate candies had a higher, statistically significant average `winpercent` than fruity candies. The correlation analysis supports this by showing that chocolate is positively correlated with `winpercent` and `pricepercent`, while `fruity` is negatively correlated with these variables, showing that fruity candies are generally less popular. The PCA results reflects this pattern by showing that the primary axis (PC1) separates chocolate/bar/popular candies from fruity/hard candies, revealing the same relationships observed in the correlation matrix. Together, the visualizations, correlation analysis, and PCA all point to the same conclusion that candies that are chocolate-based and bar-type are more likely to be popular, while fruity and hard candies tend to be less favored.